

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q20-1602

Medora 138 kV
September 19, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1602. The Interconnection Customer proposes a 164.8 MW solar photovoltaic facility interconnecting at the Medora 138 kV substation in the Pennsylvania area. The request was received on October 30, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 613 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$147,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: DELCO - WILMERDING 138 kV
Contingency: MEDORA - P.T.S 345 kV
Loading: 104.8% (Rating: 221 MVA)
DFAX: 7.6%
Pre-project loading: 99.1%
2. Facility: PARLEYS SUBSTATION - MANATI 8401 138 kV Line
Contingency: P.T.S - MEDORA 345 kV
Loading: 100.6% (Rating: 316 MVA)
DFAX: 12.0%
Pre-project loading: 94.3%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
DELCO - WILMERDING 138 kV Line	Reconductor line	\$328,000	23% (\$75,000)
PARLEYS SUBSTATION - MANATI 8401 138 kV Line	Reconductor line	\$457,000	16% (\$72,000)
	Total allocated to project		\$147,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.